

Homework (Carolina Reaper)

- Q1.** A canvas has an aspect ratio of $1 : (\sqrt{5} - 1)$. To normalize the width, find the value of $\frac{4}{\sqrt{5}-1}$.
- (A) $\sqrt{5} + 1$
(B) $4\sqrt{5} + 4$
(C) $\frac{\sqrt{5}+1}{2}$
(D) $\sqrt{5} - 1$
(E) $\frac{\sqrt{5}-1}{4}$
- Q2.** The frequency ratio between two musical notes is given by $\frac{\sqrt{3}}{\sqrt{6}+\sqrt{2}}$. Rationalize the expression.
- (A) $\frac{3\sqrt{2}-\sqrt{6}}{2}$
(B) $\frac{3\sqrt{2}-\sqrt{6}}{4}$
(C) $\frac{\sqrt{18}+\sqrt{6}}{8}$
(D) $\frac{\sqrt{2}-\sqrt{6}}{4}$
(E) $\frac{3\sqrt{2}+\sqrt{6}}{4}$
- Q3.** A sculpture's volume is scaled by a factor of $\frac{1}{\sqrt[3]{4}}$. Rationalize the scale factor.
- (A) $\frac{\sqrt[3]{2}}{4}$
(B) $\frac{\sqrt[3]{16}}{4}$
(C) $\frac{\sqrt[3]{2}}{2}$
(D) $\frac{2}{4}$
(E) $\frac{\sqrt[3]{4}}{2}$
- Q4.** A rhythm pattern duration is defined as $T = \frac{2+\sqrt{2}}{3-\sqrt{8}}$. Simplify this duration.
- (A) $10 + 7\sqrt{2}$
(B) $6 + 4\sqrt{2}$
(C) $10 - 7\sqrt{2}$
(D) $14 + 9\sqrt{2}$
(E) $10 + 3\sqrt{2}$
- Q5.** The light intensity on a mural at distance d is $\frac{12}{\sqrt{7}-\sqrt{3}}$. Simplify the intensity value.
- (A) $3\sqrt{7} - 3\sqrt{3}$
(B) $4\sqrt{7} + 4\sqrt{3}$
(C) $\frac{12(\sqrt{7}+\sqrt{3})}{10}$
(D) $3\sqrt{7} + 3\sqrt{3}$
(E) $\sqrt{7} + \sqrt{3}$
- Q6.** In a string instrument, tension differential is expressed as $\frac{\sqrt{x}}{\sqrt{x}-\sqrt{y}}$. Find the rationalized form.
- (A) $\frac{x+\sqrt{xy}}{x-y}$
(B) $\frac{x-\sqrt{xy}}{x-y}$
(C) $\frac{x+y}{x-y}$
(D) $\frac{\sqrt{x^2+xy}}{x-y}$
(E) $\frac{x+\sqrt{xy}}{x+y}$
- Q7.** A vanishing point in perspective drawing is at $P = \frac{1}{2\sqrt{3}+3\sqrt{2}}$. Rationalize P .
- (A) $\frac{3\sqrt{2}-2\sqrt{3}}{6}$
(B) $\frac{2\sqrt{3}-3\sqrt{2}}{6}$
(C) $\frac{3\sqrt{2}+2\sqrt{3}}{6}$
(D) $\frac{\sqrt{6}}{6}$
(E) $\frac{3\sqrt{2}-2\sqrt{3}}{30}$
- Q8.** The area of a mosaic tile is $\frac{\sqrt{15}}{\sqrt{5}+\sqrt{3}}$. Simplify the expression.
- (A) $\frac{5\sqrt{3}-3\sqrt{5}}{2}$
(B) $\frac{5\sqrt{3}+3\sqrt{5}}{2}$
(C) $\frac{\sqrt{75}-\sqrt{45}}{8}$
(D) $\sqrt{3} - \sqrt{5}$
(E) $\frac{15(\sqrt{5}-\sqrt{3})}{2}$

Open Ended Questions (FRQ)

- Q9.** A synthesizer modulates a signal by a factor of $M = \frac{1}{\sqrt[4]{2}}$. Rationalize the denominator to find the exact modulation factor.
- Q10.** Two stage lights create an interference pattern with intensity $I = \frac{\sqrt{a}+\sqrt{b}}{\sqrt{a}-\sqrt{b}}$. Rationalize the expression entirely given $a > b > 0$.

Chef's Tips

- Multiply by the conjugate $(a+b) \rightarrow (a-b)$ to eliminate binomial radicals.
- For n -th roots $\sqrt[n]{x^k}$, multiply by $\sqrt[n]{x^{n-k}}$ to clear the denominator.
- Check for common factors in the numerator and denominator after rationalizing to ensure simplest form.